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Language

English

TOEIC: 27th May 2024

› TOEIC : 965/990

Computer skills

Python

C/C++

Database & SQL

Matlab/Simulink

STK 12

FPGA (Quartus)

IDM

CATIA V5 : 3D
Modeling

Skills

Teamwork

Sense of initiatives
and responsibility

Strengths

Willing, Helpful,
Persevering

Travel

Austria, Morocco,
Spain, England,
Germany,
Martinique,
Guadeloupe, Greece,
Thailand, Laos
(alone), Malaysia,
Singapore,
Philippines,
Indonesia, Vietnam

Hugo LERAY

Space Systems Engineer

Space System Engineer, currently working as Research Engineer on ionospheric characterization and space weather. Experienced in multi-receiver GNSS data analysis, EPB detection and velocity estimation using ML and signal processing methods. Skilled in satcom system design, deep learning signal analysis, Python/MATLAB comms-chain development and STK mission planning. Interested in R&D roles in NewSpace, navigation, and deep tech.

Education

Master: Aeronautical and Space Systems Specialist Engineering program, English-speaking section

From September 2020 to February 2025 **Institut Polytechnique des Sciences Avancées (IPSA)** Ivry-sur-Seine, France

Space, Launchers and Satellite major.

Some projects :

- Autonomous exploration of robots using Gazebo
- Creation of a space mission to Neptune using STK and manual calculations
- Machine learning project of a character learning how to survive on an island using DEEP QL
- Creation of a full communication chain from a satellite to a receiver. Transmitter - Channel - Receiver using Python.

Semester abroad

From October 2023 to February 2024 **Universiti Sains Malaysia** Penang, Malaysia

- Satellite communication
- Antenna and propagation
- Information system theory and management

Work experience

Research Engineer - Promoted from Intern (April 2025 - September 2025)

From October 2025 to September 2026 **National University of Singapore** Singapore

Ionospheric topics, Equatorial Plasma Bubbles, GNSS signal analysis:

- Conduct research on equatorial plasma bubbles and ionospheric irregularities
- Perform multi-year GNSS data analysis to uncover novel trends and findings
- Analyze signal degradation and scintillation effects
- Develop and implement new methodologies to characterize the phenomenon
- Contribute to scientific papers and conference publications
- Collaborate with researchers on data processing and interpretation

Electromechanical technician internship

From June 2024 to August 2024 **Parker Meggitt Avrillé (Artus SAS)** Avrillé, France

Internship in the IPSA curriculum. Building mechanical, micromechanical, and electromechanical motors and gear reducers for the aerospace and defense industry.

Publications & Conferences

EuCAP 2026 (Dublin, Ireland) - Automatic Detection of Equatorial Plasma Bubbles Using Rate of TEC Index from SiReNT GNSS Data

- Neyman-Pearson detector with deep learning signal decomposition (sum of Gaussians)
- EPB clustering via IoU on Gaussian envelopes

APS-URSI 2026 (Detroit, USA) - Estimation of Equatorial Plasma Bubble Drift Velocities Using Dynamic Time Warping

- Constrained DTW alignment across receivers; drift velocity from planar wavefront model
- Validated on synthetic EPB simulator

URSI-GASS 2026 (Krakow, Poland) - Equatorial Plasma Bubble Velocity Estimation over Southeast Asia

- DTW-based velocity framework on SiReNT data with spatial interpolation